Chao Yang

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Working Experience

Shanghai AI Lab — **Shanghai, China**
Postdoctoral Researcher, Leading large model safety & decision group — *April. 2022 – Current*

- Alignment and Safety of Large Language Model, Multi-modal LLM.
- Robotic Manipulation and Embodied Intelligence.
- Co-workers: Prof. Yu Qiao, Dr. Jing Shao, Dr. Yu Liu.

Education Experience

Tsinghua University — **Beijing, China**
Aug.2018 – July.2022

- Ph.D. Student, In Department of Computer Science and Technology
- Advisors: Prof. Fuchun Sun

University of Hamburg — **Hamburg, Germany**
Aug.2019 – Sep.2019

- Visiting Student, In UHH · Department of Informatics
- Advisors: Prof. Jianwei Zhang

Tsinghua University — **Beijing, China**
Sep.2015 – Jun.2018

- M.Eng, In Computer Science and Technology
- Advisors: Prof. Fuchun Sun

Sichuan University — **Chengdu, China**
Sep.2011 – Jun.2015

- B.Eng, In Electronics and Information Engineering
- Advisors: Prof. Qingchuan Tao

Research Interests

My research interest includes Large Language Model Safety, Multi-modal Large Model, and Robotic Embodied Intelligence. Some of my current research keywords can be found below:

- **Large Language Model**: LLM Alignment and Fine-tuning, LLM Attack and Defense.
- **Multi-modal LLM**: Vision and language Fusion, Video-QA, VQA.
- **Embodied Robotics**: Robotic Manipulation, Reinforcement and Imitation Learning.

Manuscripts

Weak-to-Strong Search: Align Large Language Models via Searching over Small Language Models

- Zhanhui Zhou, Zhixuan Liu, Jie Liu, Zhichen Dong, Jiaheng Liu, Chao Yang‡, Yu Qiao. UNDER REVIEW
- https://arxiv.org/abs/2405.19262

GaussianGrasper: 3D Language Gaussian Splatting for Open-vocabulary Robotic Grasping.

- tba

LVLM-SafeRR: Benchmarking Safety-guided Reasonable Refusal in Large Vision-Language Models

- tba

Guided Sequence Generation for Inference Time Alignment

- tba

TreeEval: Benchmark-Free Evaluation of Large Language Models through Tree Planning

- tba

Publications (Google Scholar¶)

*(equal contribution), ‡(corresponding author)

Emulated Disalignment: Safety Alignment for Large Language Models May Backfire!

- Zhanhui Zhou, Jie Liu, Zhichen Dong, Jiaheng Liu, Chao Yang‡, Yu Qiao.
- In The 62nd Annual Meeting of the Association for Computational Linguistics (ACL2024 Main)

SEER: Facilitating Structured Reasoning and Explanation via Reinforcement Learning

- Guoxin Chen, Kexin Tang, Chao Yang‡, Yu Qiao.
- In The 62nd Annual Meeting of the Association for Computational Linguistics (ACL2024 Main)

Beyond one-preference-for-all: Multi-objective direct preference optimization

- Zhanhui Zhou, Jie Liu, Chao Yang‡, Yu Qiao.
- In The 62nd Annual Meeting of the Association for Computational Linguistics (ACL2024 Findings)

RoboCodeX: Multimodal Code Generation for Robotic Behavior Synthesis

- Yao Mu, Junting Chen, Qinglong Zhang, Shoufa Chen, Qiaojun Yu, Chongjian Ge, Runjian Chen, Zhixuan Liang, Mengkang Hu, Chaofan Tao, Peize Sun, Haibao Yu, Chao Yang, Wenqi Shao, Wenhai Wang, Jifeng Dai, Yu Qiao, Mingyu Ding, Ping Luo.
- In The Forty-first International Conference on Machine Learning (ICML2024)

Safety of Multimodal Large Language Models on Images and Text

- Xin Liu, Yichen Zhu, Yunshi Lan, Chao Yang‡, Yu Qiao
- In *International Joint Conference on Artificial Intelligence (IJCAI 2024 Survey Track)*.

Attacks, defenses and evaluations for llm conversation safety: A survey

- Zhichen Dong*, Zhanhui Zhou*, Chao Yang‡, Jing Shao, Yu Qiao
- In *2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2024)*.

LLaMA-Excitor: General Instruction Tuning via Indirect Feature Interaction

- Chao Yang*, Bo Zou*, Yu Qiao, Chengbin Quan, Youjian Zhao
- In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024)*.

VideoDistill: Language-aware Vision Distillation for Video Question Answering

- Chao Yang*, Bo Zou*, Yu Qiao, Chengbin Quan, Youjian Zhao
- In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024)*.

Critic-guided decision transformer for offline reinforcement learning

- Chao Yang*, Yuanfu Wang*, Ying Wen, Yu Liu, Yu Qiao
- In *AAAI Conference on Artificial Intelligence (AAAI 2024)*.

Rethinking Mutual Information for Language Conditioned Skill Discovery on Imitation Learning

SpaceCLIP: A Vision-Language Pretraining Framework With Spatial Reconstruction On Text

- Chao Yang*, Bo Zou*, Chengbin Quan, Youjian Zhao
- In *Proceedings of the 31st ACM International Conference on Multimedia (ACM-MM 2023)*.

Sim2Real Object-Centric Keypoint Detection and Description

- Chao Yang*, Chengliang Zhong*, Fuchun Sun, Xiaodong Mu, Wenbing Huang.
- In *AAAI Conference on Artificial Intelligence (AAAI 2022)*.

Evaluations of the Gap between Supervised and Reinforcement Lifelong Learning on Robotic Manipulation Tasks

- Fan Yang, Chao Yang, Huaping Liu, Fuchun Sun.
- In *5th Conference on Robot Learning (CoRL 2021)*.

Graph Topography-Aware Reinforcement Learning for Intelligent Traffic Signal Control

- Chao Yang, Bo Zou, Wenbing Huang, Fuchun Sun, Huaping Liu.
- in *IEEE Int. Conf. on CYBER Technology in Automation, Control, and Intelligent Systems.(IEEE-Cyber 2021)*. ***(Finalist Best Poster Award)***

Fault-Aware Robust Control via Adversarial Reinforcement Learning

- Fan Yang, Chao Yang, Di Guo, Huaping Liu, Fuchun Sun.
- in *IEEE Int. Conf. on CYBER Technology in Automation, Control, and Intelligent Systems.(IEEE-Cyber 2021)*. ***(Best Poster Award)***

Adversarial Skill Learning for Robust Manipulation

- Chao Yang*, Pingcheng Jian*, Di Guo, Huaping Liu, Fuchun Sun
- In *IEEE International Conference on Robotics and Automation.(ICRA 2021)*.

Visual-Tactile Fusion for Robotic Stable Grasping

- Bin Fang, Chao Yang*, Fuchun Sun, Huaping Liu
- In *Industrial Robotics-New Paradigms* 2020.

Multi-Modal Continual Learning using Online Dictionary Updating

- Fuchun Sun, Huaping Liu, Chao Yang, Bin Fang
- In *IEEE Transactions on Cognitive and Developmental Systems* 2020.

Reinforcement Learning from Imperfect Demonstrations under Soft Expert Guidance

- Mingxuan Jing, Xiaojian Ma, Wenbing Huang, Fuchun Sun, Chao Yang, Bin Fang, Huaping Liu
- In *AAAI Conference on Artificial Intelligence.(AAAI 2020)*.

Imitation learning from observations by minimizing inverse dynamics disagreement

- Chao Yang*, Xiaojian Ma*, Wenbing Huang, Fuchun Sun, Huaping Liu, Chuang Gan, Junzhou Huang
- In *Advances in Neural Information Processing Systems.(NeurIPS 2019)*. **(Spotlight Oral)**.

Multimodal grasp dataset: A novel visual–tactile data set for robotic manipulation

- Tao Wang, Chao Yang, Frank Kirchner, Peng Du, Fuchun Sun, Bin Fang
- In *International Journal of Advanced Robotic Systems* 2019.

Predict Robot Grasp Outcomes based on Multi-Modal Information

- Chao Yang, Peng Du, Fuchun Sun, Bin Fang, Jie Zhou
- In *IEEE International Conference on Robotics and Biomimetics.(ROBIO 2018)*.

A dual-modal vision-based tactile sensor for robotic hand grasping

- Bin Fang, Fuchun Sun, Chao Yang‡, Hongxiang Xue. (‡Corresponding author)
- In *IEEE International Conference on Robotics and Automation.(ICRA 2018)*.

Robotic grasping using visual and tactile sensing

- Di Guo, Fuchun Sun, Bin Fang, Chao Yang, Ning Xi
- In *Information Sciences* 2017.

A Vision-Based Tactile Sensor with Tactile Detection Using Neural Network

- Chao Yang, Fuchun Sun, Bin Fang, Luxuan Li.
- In *International Conference on Cognitive Systems and Signal Processing.(ICCSIP 2016).*

Workshop

Rethinking Mutual Information for Language Conditioned Skill Discovery on Imitation Learning.
Rethinking Mutual Information for Language Conditioned Skill Discovery. (Link)

- Chao Yang, Zhaoxun Ju. *RSS 2023 Workshop: Articulate Robots: Utilizing Language for Robot Learning.*

RGB-D Object Segmentation for Multi-Step Pick-and-Place in Open Cloud Robot Table

- Chao Yang, Chengliang Zhong, Mingxuan Jing, Yu Luo, Tianying Ji, Yikai Wang, Wenbing Huang, Xiaodong Mu, Fuchun Sun. *ICRA 2021 Workshop: Cloud-Based Competitions and Benchmarks for Robotic Manipulation and Grasping.*

Internship Experience

Xiaomi AI Lab — **Beijing, China**
Research Intern — *Dec. 2019 – Jun. 2020*

- Research on Generative Adversarial Networks and Domain Adaptation.
- Mentor: Dr. Deli Zhao

Tencent AI Lab — **ShenZhen, China**
Research Intern — *Aug. 2018 – Jun. 2019*

- Research on Reinforcement Learning and Imitation Learning from Observation.
- Mentor: Dr. Wenbing Huang

Honers and Awards

Shanghai Postdoctoral Excellent Program — **Shanghai, China**
- For our research on multi-modal perception and robotic skill learning. — *Dec. 2022*

'84' Future Innovation Scholarship (RMB 100,000) — **Tsinghua, China**
- For our research on multi-modal perception and robotic skill learning. — *Dec. 2020*

Open Cloud Robot Table Organization Challenge(OCRTOC) — **Online, China**
- **First place** in simulation stage and third place in real stage. — *Nov. 2020*

Robotic Grasping and Manipulation Competition@IROS 2019 — **Macau, China**
- **First place** in robotic logistics track. — *Nov. 2019*

Robotic Grasping and Manipulation Competition@IROS 2017 — **Vancouver, Canada**
- **Third place** in robotic service track. — *Nov. 2017*

Robotic Grasping and Manipulation Competition@IROS 2016 — **Daejeon, Korea**
- **First place** in robotic automatic pick-and-place track. — *Nov. 2016*

Excellent Undergraduate Student **Chengdu, China**

- For their excellent performance during the four years of college life. *Jun. 2015*

China National Scholarship (Best Undergraduate Scholarship) **Chengdu, China**

- Reward the top 1% students during undergraduate study (every year). *Oct.2012 – Oct.2014*

Academic Services/Teaching

- Conference Reviewer: IROS20-24, ICRA21-23, CoR21-23, ICLR22-24, NIPS20-24, CVPR23
- Journal Reviewer: IEEE Robotics and Automation Letters (RA-L).
- Head TA and Part-time Instructor: Intelligent Computing and Robot Learning 2018-2021; Artificial Intelligence Theory 2020; System Analysis and Control 2019-2021 in Tsinghua University.